HARDWARE LABORATORY 2

TECHNOLOGIES FOR IOT ECOSYSTEMS

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EXERCISE 1

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Exercise 1

Specification 1)

To reduce the number of wires and connections, you can also use the internal temperature sensor of the Arduino…

Although less precise and accurate than the external one.

Board Temperature Sensor

The board includes a LSM6DSOXTR Inertial Measurement Unit (IMU) from STM.

3D Accelerometer + 3D Gyroscope + Temperature Sensor

We can use the IMU functionality by including the corresponding library:

```
#include <Arduino_LSM6DSOX.h>
```

Board Temperature Sensor

In the setup() we must initialize the library as follows:

```
if (!IMU.begin()) {
  Serial.println("Failed to initialize IMU!");
  while(1);
}
```

Lastly, we can read the temperature (directly in Celsius, as an int) with the following code:

Drawback: the resolution is 1 degree, and the sensor less accurate than the external one.

```
if (IMU.temperatureAvailable()) {
  int temperature = 0;
  IMU.readTemperature(temperature);
}
```

Exercise 1

Specification 2)

LED light is proportional to current

With the circuit of Exercise 1.1, we can regulate the current by reducing the voltage drop.

We can control LED intensity using PWM!

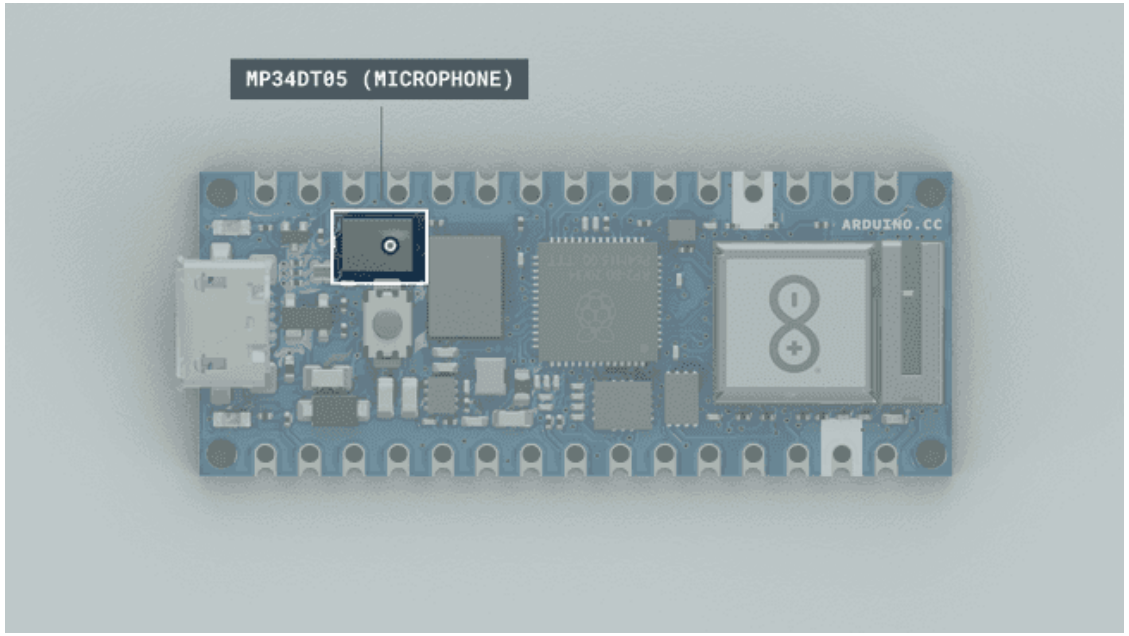
Exercise 1

Specification 4)

The RP2040 Connect has an on-board MP35DT05 microphone that uses PDM (Pulse-Density Modulation)

PDM is a more general case of PWM with variable period.

Datasheet.



Exercise 1

Luckily the PDM arduino library does the PDM-to-digital conversion for you

You should use this library to interact with the microphone

Include the header file:

```
#include <PDM.h>
```

In the globals, define a buffer to store the audio samples (converted to 16bit digital values). By default, the library expects a 512 bytes buffer. This can be changed with (setBufferSize()):

```
// Buffer to read samples into, each sample is 16-bits
volatile short sampleBuffer[256];
```

Exercise 1

In the setup(), associate a callback to execute everytime new data arrives

Callback function (ISR)

```
PDM.onReceive(onPDMdata);
```

Then initialize the PDM sensor:

N. Channels (1 = mono) **Sample frequency**

```
if (!PDM.begin(1, 16000)) {
  Serial.println("Failed to start PDM!");
  while (1);
}
```

Exercise 1

ISR function:

- Check how many bytes are available
- Read them in the buffer
- Remember that each sample requires 2 bytes

Note: it's an ISR, so you can't call delay(), Serial.print(), etc.

```
void onPDMdata() {
  // Query the number of available bytes
  int bytesAvailable = PDM.available();
  // Read into the sample buffer
  PDM.read(sampleBuffer, bytesAvailable);
  // 16-bit, 2 bytes per sample
  samplesRead = bytesAvailable / 2;
}
```

Exercise 1

In the loop(), use the buffer values as needed…

```
void loop() {
  if (samplesRead) {
    for (int i = 0; i < samplesRead; i++) {
      Serial.println(sampleBuffer[i]);
    }
    samplesRead = 0;
  }
}
```

Note: this is an example, implementing the lab request is more tricky!!

Exercise 1

Specification 6)

When presence is detected:

$T_{AC,min} = T_{AC,min,pres}$
$T_{AC,max} = T_{AC,max,pres}$

etc. (same for heater)

Otherwise:

$T_{AC,min} = T_{AC,min,abs}$
$T_{AC,max} = T_{AC,max,abs}$

etc. (same for heater)

Exercise 1

The rest is up to you…

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